fit for commercial purpose. This unit is not always necessary.

8. Overhead Reservoir

The function of this unit is to store the purified water after the treatment is complete. The water from the reservoir is supplied to the consumers by gravity or pumping.

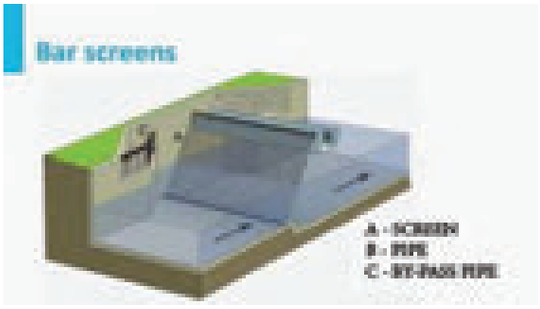
Location of Treatment units

The following points are to be considered while locating the plants for treatment of water

1. All plants should be located in such a sequence that water may flow from one process to the other automatically.
2. Elevation of different processes should ensure gravity flow from one to another.
3. All the processing units should require minimum area. For future expansion, adequate space should be available.
4. All the processing units should be nearer to the area of distribution for better working and efficient maintenance.
5. The site of treatment plant installation should be neat and clean and should be well protected.
6. A well equipped laboratory should be available at the treatment plant to have regular checking of the quality.
7. Space for the disposal of sludge from settling tanks and wash water from filters should be provided.

4.4.2 Screening

Water, when derived from the surface sources may contain floating



Bar screens

matters. Most of the big and visible objects such as trees, branches, sticks, vegetation, fishes, dead and floating animals, etc., can be removed by screening. (e.g.) Bar screens. Bar screens may be of two types.

1. Coarse screens
2. Fine screens

4.4.3 Sedimentation

It is the process of causing heavier solid particles in suspension both organic and inorganic, to settle by retaining the water in a huge tank called sedimentation tank.

Theory of Sedimentation

The particles heavier than water tend to settle down due to the force of gravity. Impurities in water are held in suspension due to the turbulence of the moving water. When this turbulence is checked and the velocity of flow is reduced, the suspended particles tend to settle down at the bottom of the tank. The settling velocity depends upon

1. The horizontal velocity of flow.
2. The shape and size of the particle.
3. The specific gravity of the particle.
4. The temperature of water.

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